Pavel Tyapkin

Senior FPGA / DSP Engineer | SDR & Wireless Communications

Ph.D. equivalent (Candidate of Technical Sciences), Radio Engineering | Currently based in Latin America | Open to relocation / remote
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Professional Profile

FPGA and digital signal processing engineer with approximately 10 years of hands-on experience in communication systems, real-time signal processing, SDR, and embedded platforms. Experienced in developing and integrating RTL for FPGA/SoC devices, optimizing FPGA designs for timing and resource utilization, implementing hardware interfaces, and validating communication algorithms through simulation and experimental testbeds.

Technical Expertise

Area	Technologies and experience
FPGA / RTL	Verilog, Xilinx Vivado and ISE, Zynq-7000, Artix-7, Intel Quartus; RTL design, integration, timing closure, resource optimization, hardware bring-up
Verification / debug	ModelSim, Xilinx ILA/VIO, simulation, on-chip debugging; basic experience with Verilator
DSP / communications	MATLAB, Simulink, GNU Octave; OFDM, BPSK/QPSK, digital filtering, multichannel reception, synchronization, signal separation, ICA and other blind signal processing techniques
SDR / RF	GNU Radio, USRP B210/N210, ADRV9009, AD9361, SDR-based measurement and experimental setups; experience evaluating mesh-radio technologies
Embedded systems	Embedded Linux, Buildroot, ARM-based platforms, hardware/software integration, register-mapped peripheral control, C/C++, Python
Interfaces / data acquisition	AXI / AXI-Lite, DMA (PL–PS), PCIe Gen3 / XDMA, Ethernet, SPI, I²C, UART, ADC/DAC interfaces
Additional tools	Qt, RFNoC, KiCad, Altium Designer (schematic review and prototyping), oscilloscopes, RF signal generators

Professional Experience

Lead Radio Engineer — AO Energiya Razvitiya (ER)

October 2025 – October 2026

- Developed, integrated, and optimized FPGA logic for IEEE 802.11-based wireless communication systems, building on established implementations and architectures.
- Performed RTL-level timing and resource optimizations, toolchain migration, and hardware adaptation across FPGA/SoC platforms.
- Developed FPGA-side control logic for RF peripherals and automatic gain regulation, with configurable software-accessible control and monitoring registers.

Project details and quantitative results are intentionally omitted for confidentiality.

FPGA / Embedded Systems Developer — NII KP, Branch of ORKK JSC

November 2020 – April 2025

- Developed and verified DSP algorithms, digital filters, and receiver/transmitter processing chains for satellite communication and COSPAS–SARSAT-related systems.
- Designed and integrated FPGA logic for multichannel data acquisition and real-time processing using Xilinx 7-series FPGA/SoC devices.
- Integrated ADC/DAC devices and digital peripherals; implemented communication and data-transfer paths using SPI, I²C, UART, PL–PS DMA, and PCIe/XDMA.

Engineer — Topcon Positioning Systems

September 2018 – September 2020

- Developed DSP algorithms and FPGA/DSP logic for navigation receiver systems.
- Designed SDR-based automated test and measurement setups for radio equipment evaluation and ETSI-related testing.
- Developed signal generation and measurement automation software for receiver/transmitter validation.

Engineering Technician — Russian Federal Space Agency

March 2016 – June 2018

- Participated in development and verification of FPGA-based satellite communication systems.
- Built simulation models to evaluate DSP algorithms and communication subsystem architectures.

- Developed hardware/software test setups for communication equipment evaluation.

Research Experience

Junior Researcher — Moscow Aviation Institute (MAI)

2023 – December 2025 · Russian Science Foundation-funded research project

- Investigated independent component analysis (ICA) and blind source separation methods for interference mitigation in deep-space communication systems.
- Designed and assembled experimental hardware prototypes and testbeds to evaluate signal processing methods under representative interference conditions.

Project: Research into improving the efficiency of deep-space communication systems using relay spacecraft.
Russian Science Foundation project information

Teaching Experience

Teaching Instructor — Moscow Aviation Institute (MAI)

2018 – June 2025

- Conducted tutorials and laboratory sessions for **General Communication Theory**; occasionally delivered lectures.

Education

Ph.D. equivalent / Candidate of Technical Sciences — Radio Engineering (2025)
Moscow Aviation Institute (MAI)
Dissertation: *Improving the Interference Immunity of Radio Communication Systems Against Impulsive Interference Using Blind Source Separation Methods.*

Postgraduate studies — Telecommunication Systems, Networks and Devices (completed 2024)
Moscow Aviation Institute (MAI)

M.Sc. — Infocommunication Technologies and Communication Systems (2020)
Moscow Aviation Institute (MAI)

B.Sc. — Infocommunication Technologies and Communication Systems (2018)
Moscow Aviation Institute (MAI)

Selected Publications and Research Outputs

- **Tyapkin, P. S., Vazhenin, N. A., Plokhikh, A. P.** “Analysis of Effectiveness of Methods of Protection against Interference Emission from Stationary Plasma Thrusters in Deep-Space Communication Systems.” *Russian Aeronautics*, 2025, 68(2), pp. 501–507.
- **Tyapkin, P. S., Vazhenin, N. A., Plokhikh, A. P.** “Improvement of Interference Immunity of Deep Space Communication Systems Exposed to Interference from Stationary Plasma Thrusters.” *Russian Aeronautics*, 2024, 67(2), pp. 427–439.

Languages

- **Russian:** Native
- **English:** Advanced (C1, self-reported)